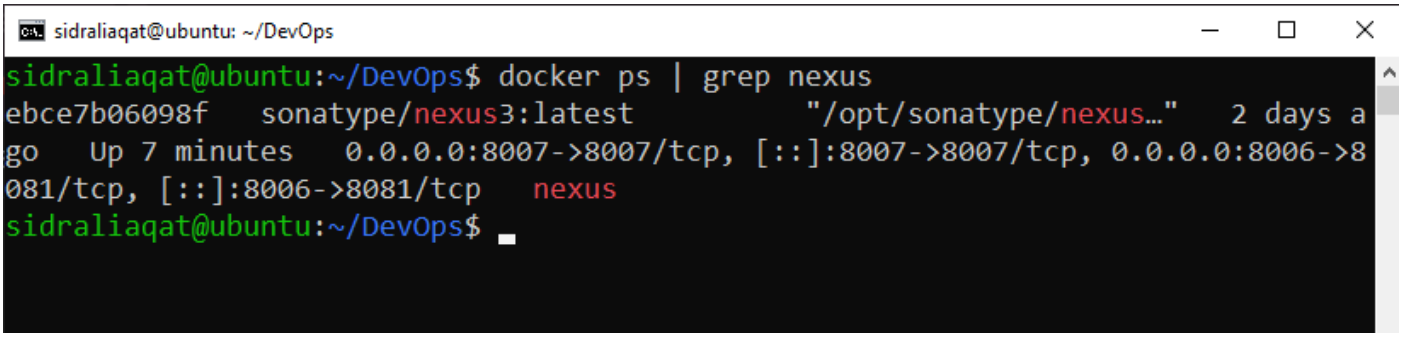
Lab 6: Set Up Nexus Repository and Configure User Access

Name	Sidra Liaqat
Reg no	2023-BCS-083
Class	BCS-7A
GitHub username	sidraliaqat
Course	DevOps (DevOps_F26)

Task 1: Run Nexus Repository



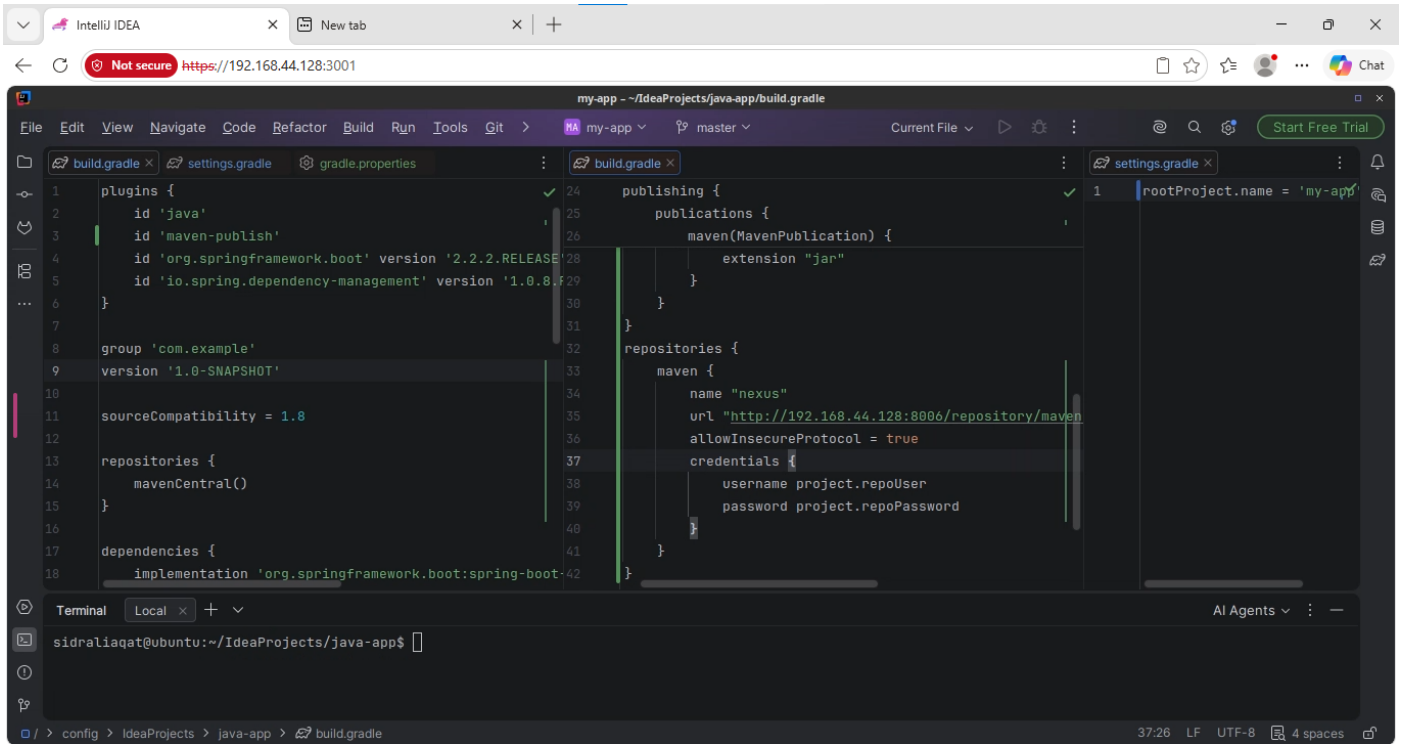
nexus_container_running.png
Nexus container running in Docker with ports 8006 and 8007 published.

Task 2: Create a Role and User and Verify Access

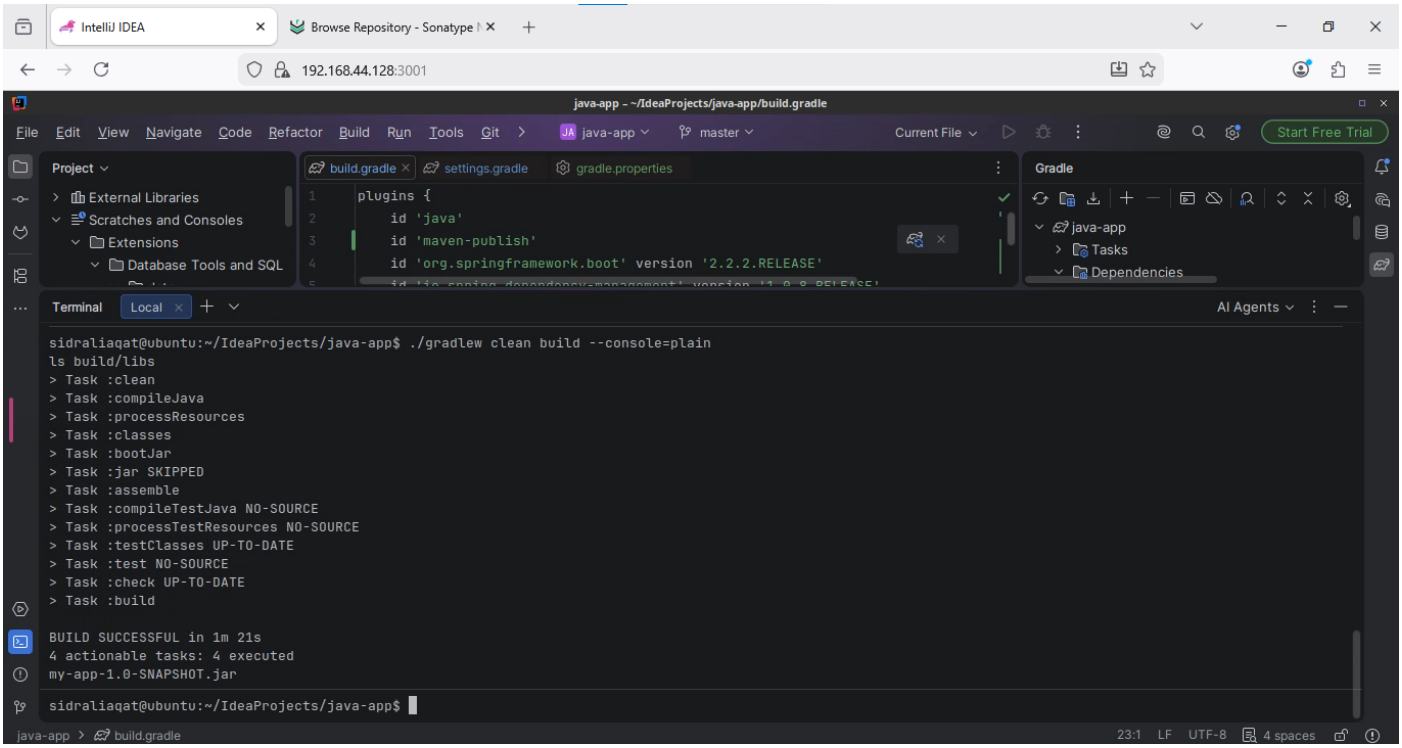
Screenshot pending: nexus_user_access.png

nexus_user_access.png
Status API request as the nexus user returns HTTP/1.1 200 OK.

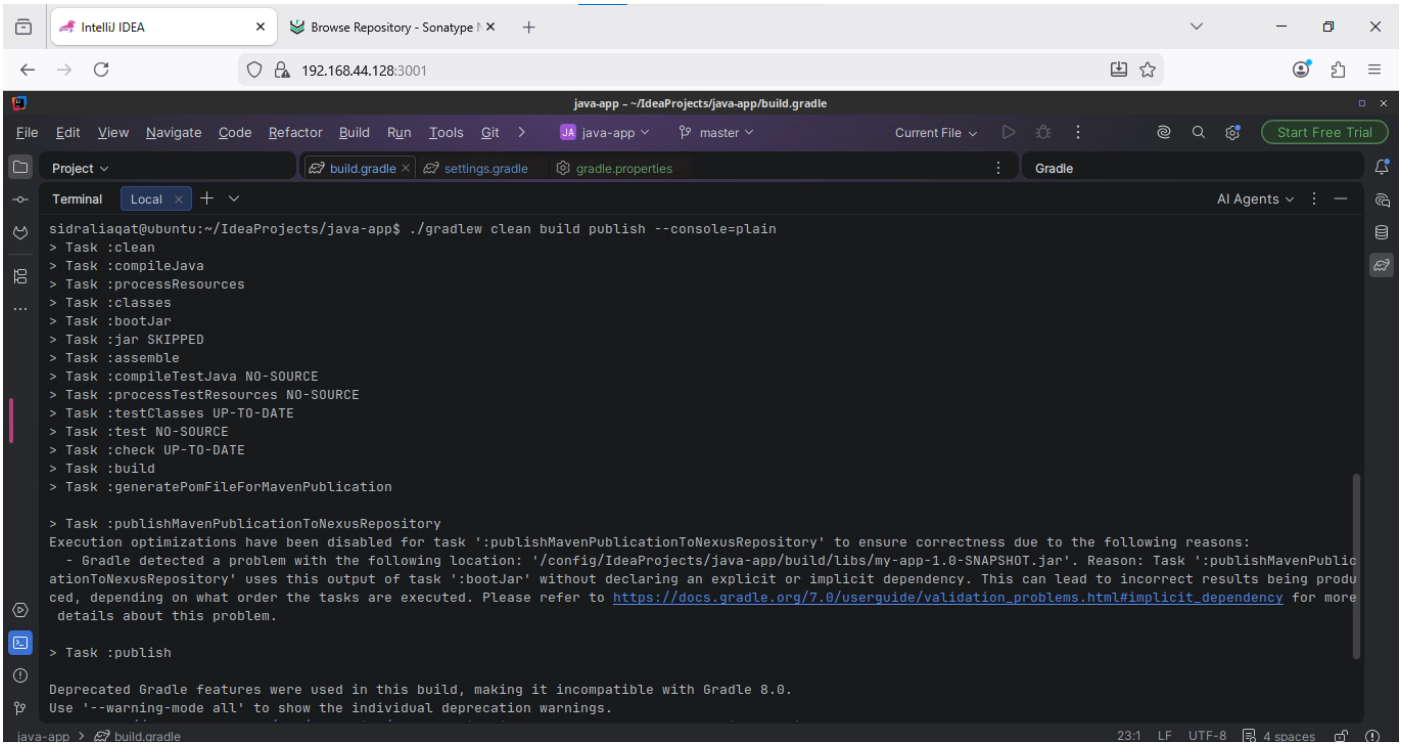
Task 3: Build and Publish the Gradle Application



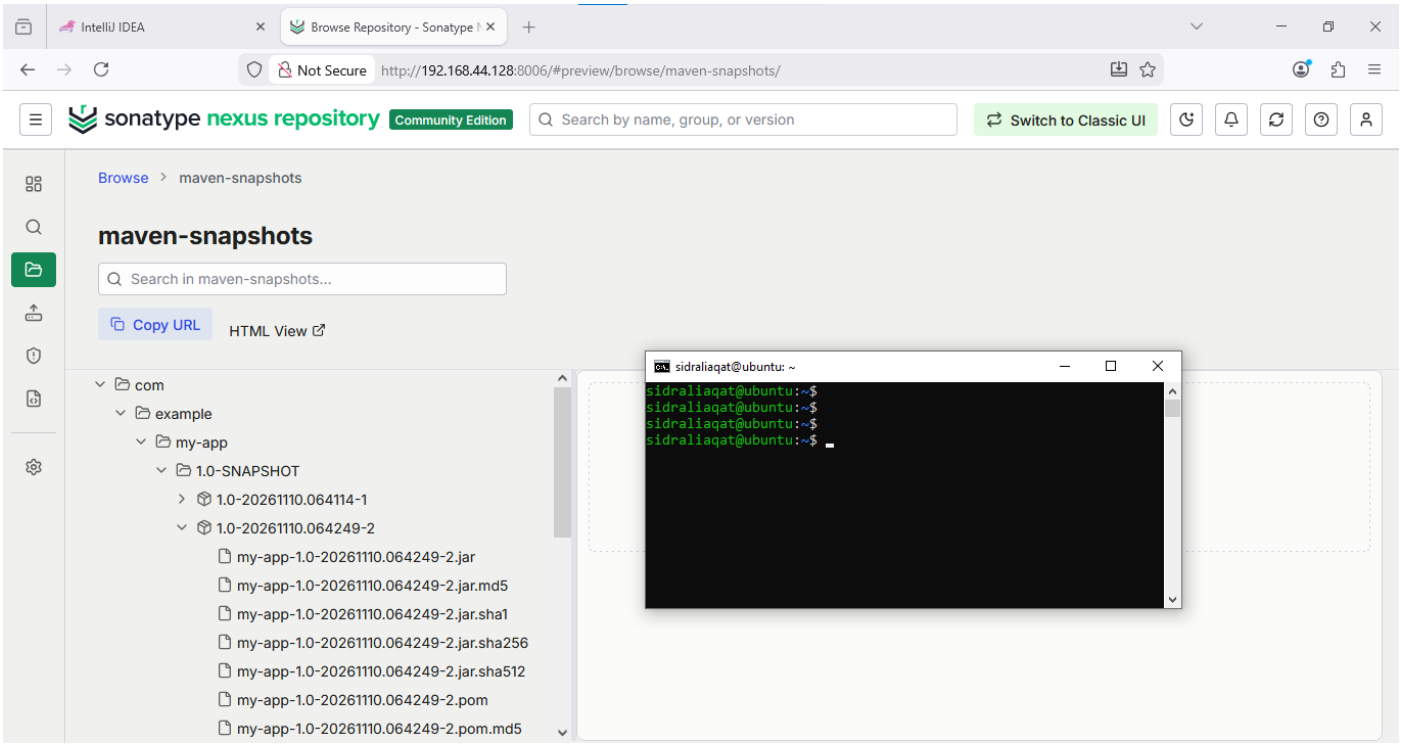
gradle_nexus_configuration.png
build.gradle publishing configuration and settings.gradle (rootProject.name = 'my-app').



gradle_nexus_build.png
./gradlew clean build succeeds and creates my-app-1.0-SNAPSHOT.jar.

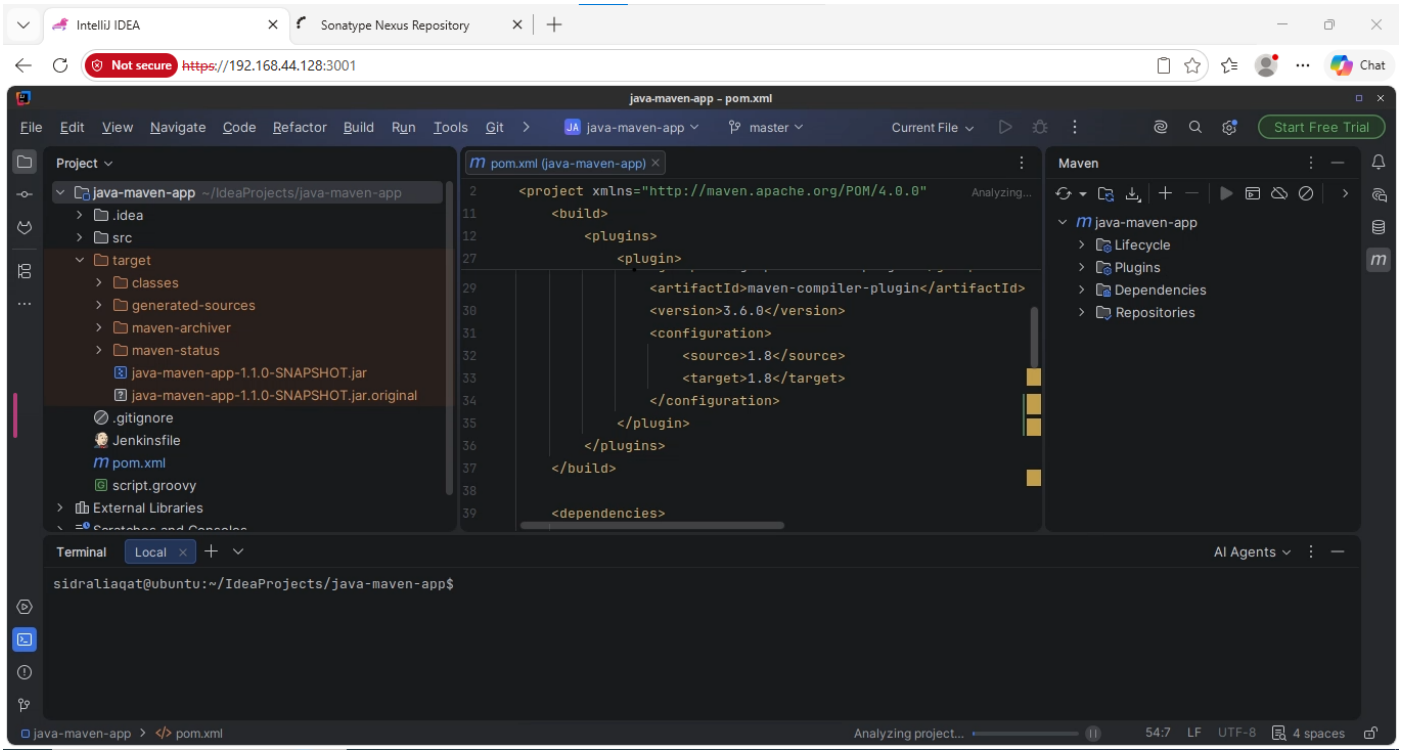


gradle_nexus_publish.png
./gradlew clean build publish runs publishMavenPublicationToNexusRepository.

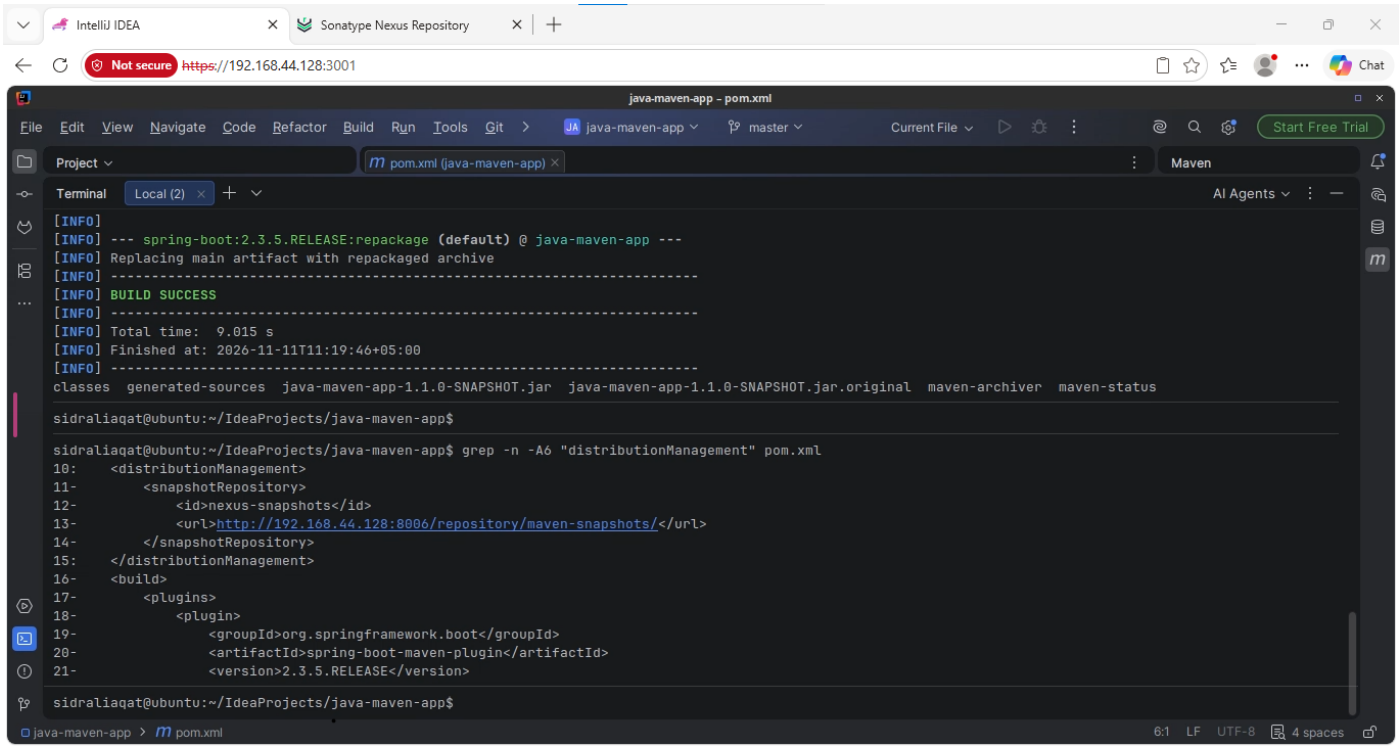


nexus_gradle_artifact.png
Published my-app snapshot and Maven metadata under com/example/my-app/1.0-SNAPSHOT.

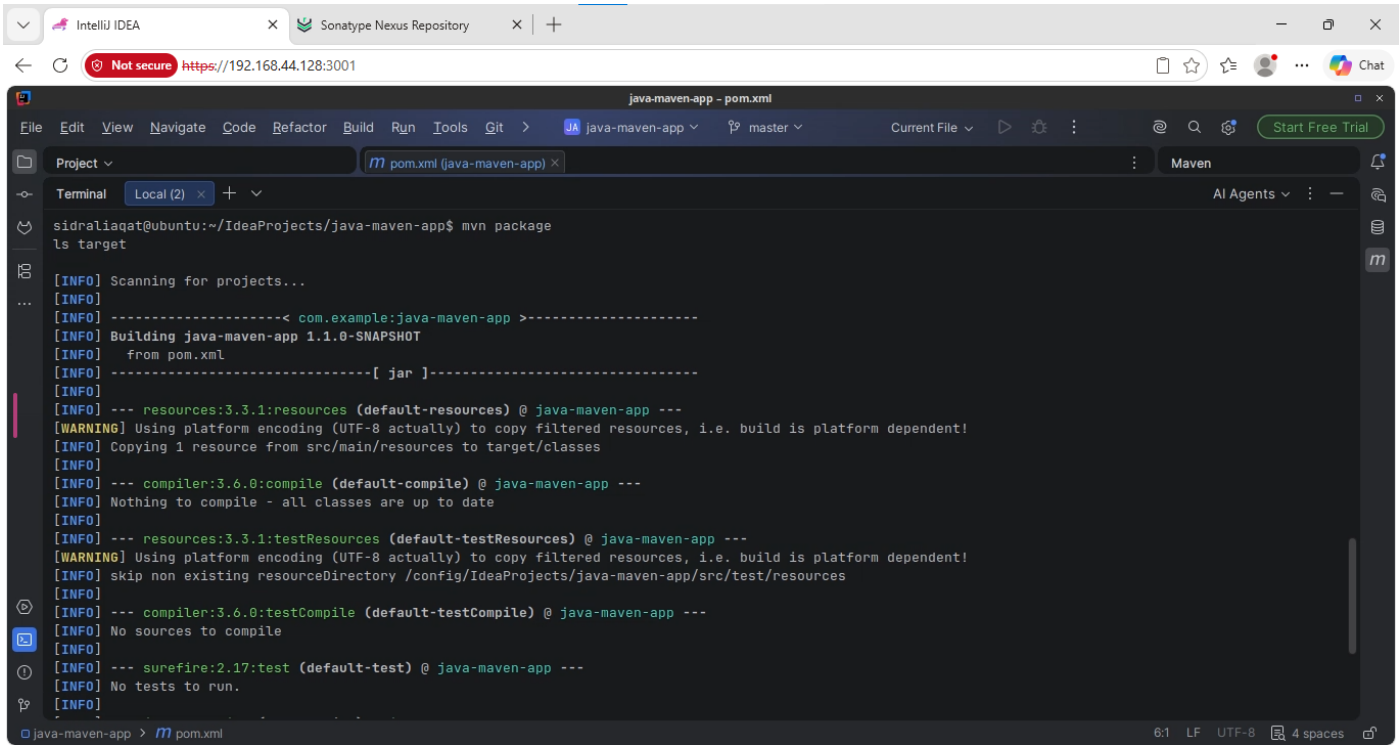
Task 4: Publish the Maven Application



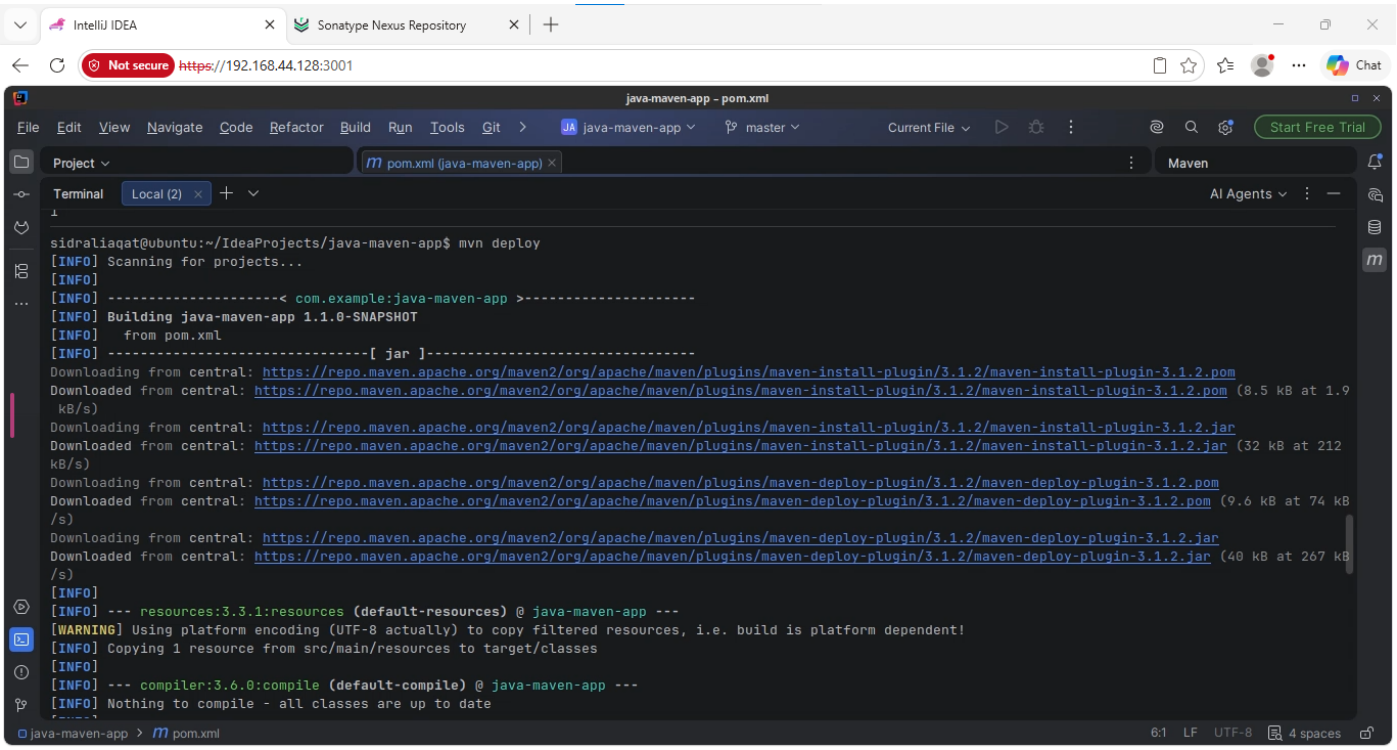
maven_nexus_project.png
java-maven-app project open in IntelliJ.



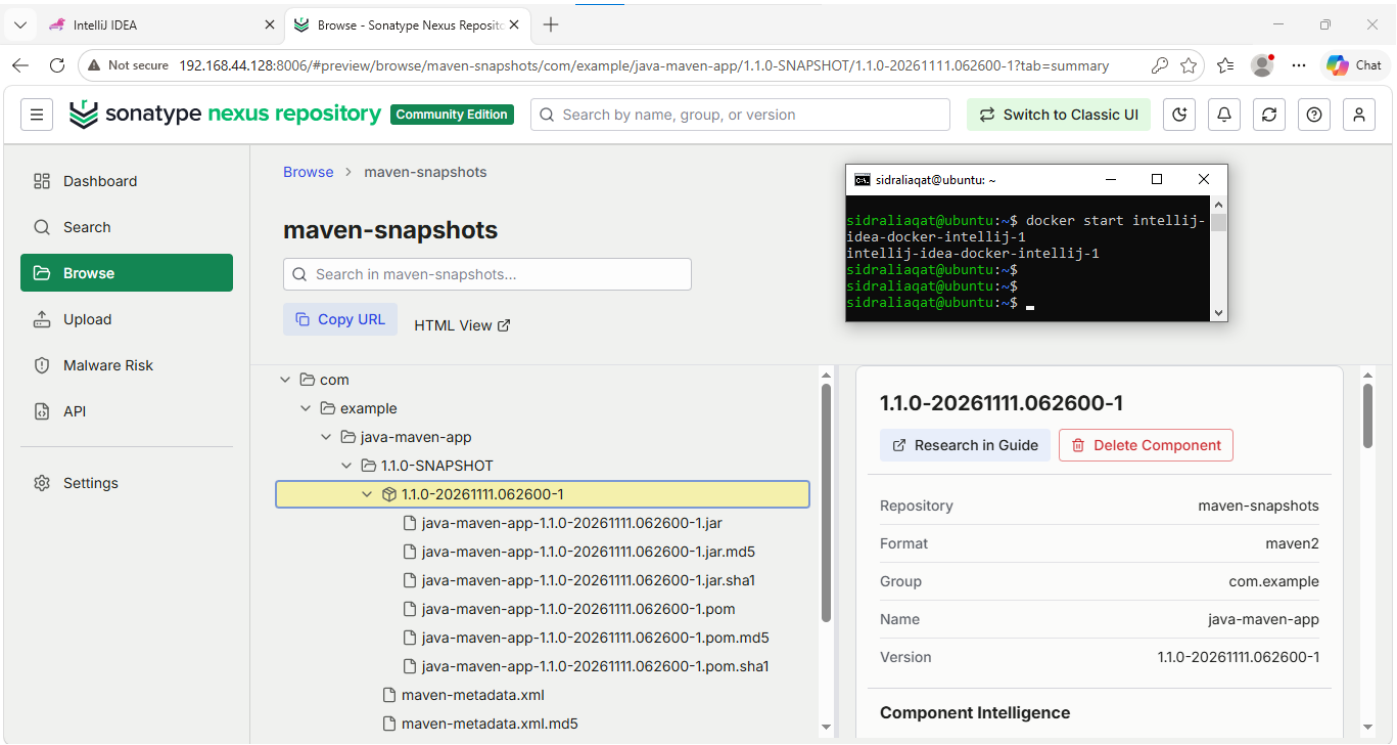
maven_distribution_management.png
distributionManagement block with the nexus-snapshots repository in pom.xml.



maven_package_nexus.png
mvn package builds the snapshot JAR.



maven_deploy_nexus.png
mvn deploy uploads the snapshot to Nexus.



nexus_maven_artifact.png
java-maven-app 1.1.0-SNAPSHOT with JAR, POM and metadata in Nexus.

Task 5: Inspect Nexus Using the REST API

```
sidraliaqat@ubuntu: ~
sidraliaqat@ubuntu:~$
sidraliaqat@ubuntu:~$
sidraliaqat@ubuntu:~$ curl -u nexus -X GET "http://192.168.44.128:8006/service/rest/v1/repositories"
Enter host password for user 'nexus':
[ {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.44.128:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
sidraliaqat@ubuntu:~$
```

nexus_repositories_user.png
Repository list returned for the restricted nexus user (maven-snapshots only).

```
sidraliaqat@ubuntu: ~
  "online" : true,
  "size" : 0,
  "attributes" : { }
sidraliaqat@ubuntu:~$ curl -u admin -X GET "http://192.168.44.128:8006/service/rest/v1/repositories"
Enter host password for user 'admin':
[ {
  "name" : "maven-releases",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.44.128:8006/repository/maven-releases",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.44.128:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-central",
  "format" : "maven2",
  "type" : "proxy",
  "url" : "http://192.168.44.128:8006/repository/maven-central",
  "online" : true,
  "size" : 0,
  "attributes" : {
    "proxy" : {
      "remoteUrl" : "https://repo1.maven.org/maven2/"
    }
  }
}, {
  "name" : "maven-public",
```

nexus_repositories_admin.png
Repository list returned for the admin user.



nexus_components_list.png
Components in maven-snapshots, showing id, repository, name and version.

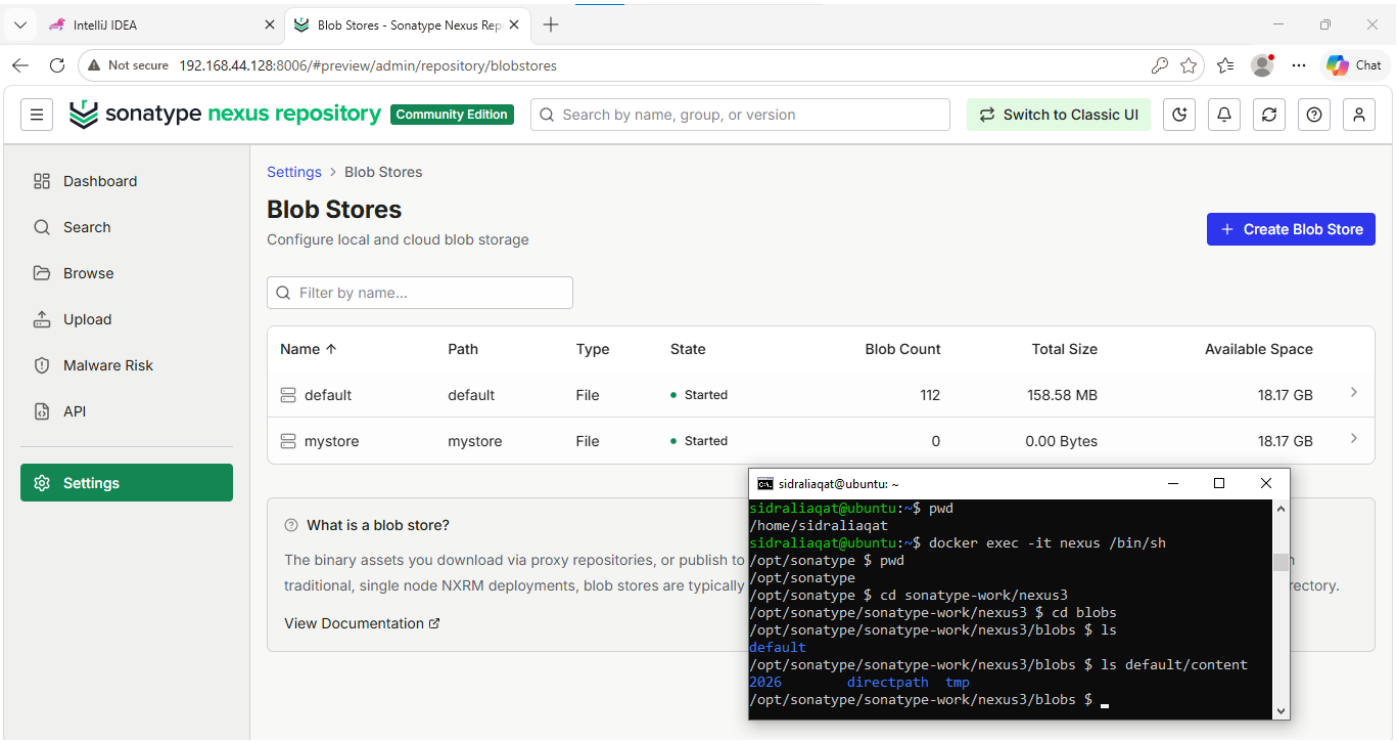


nexus_component_details.png
One component fetched by its id.

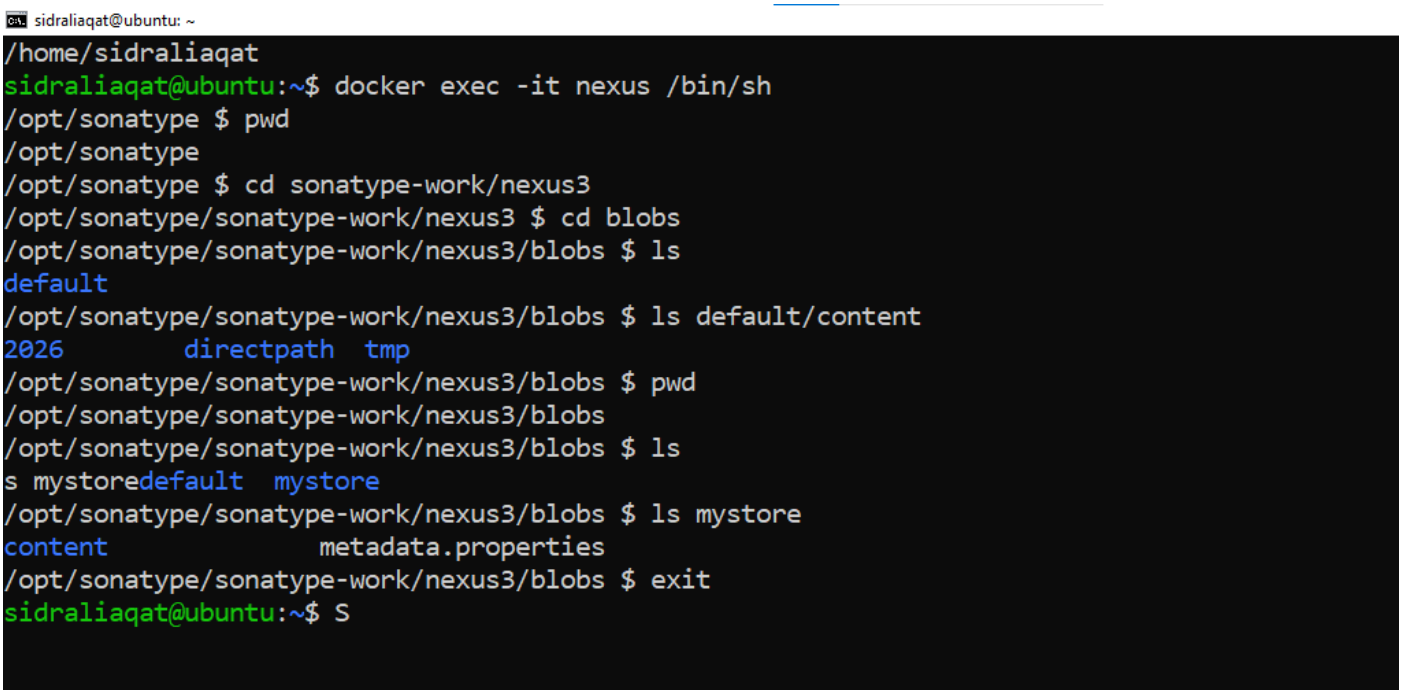
Task 6: Blob Store and Docker Hosted Repository

Screenshot pending: nexus_default_blob_content.png

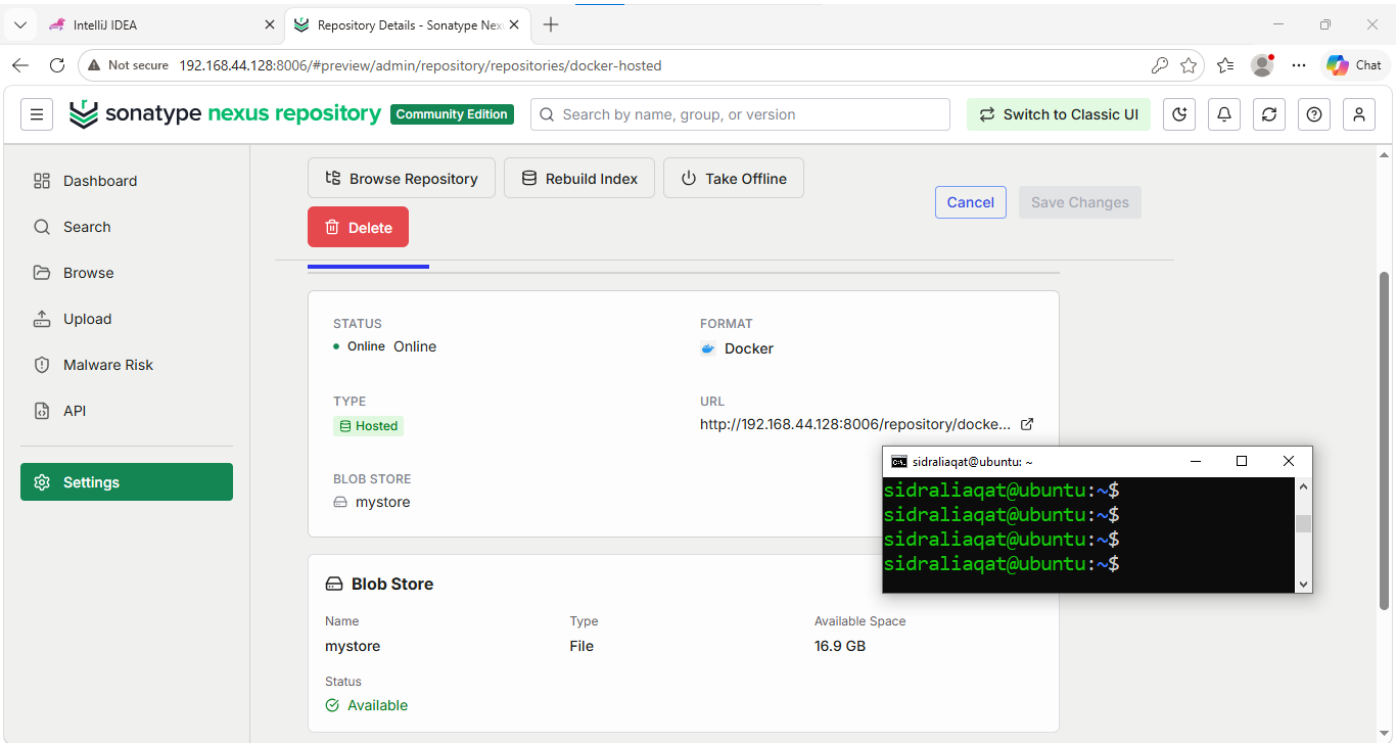
nexus_default_blob_content.png
Default blob store directory inside the Nexus container.



nexus_mystore_created.png
default and mystore file blob stores, both Started.

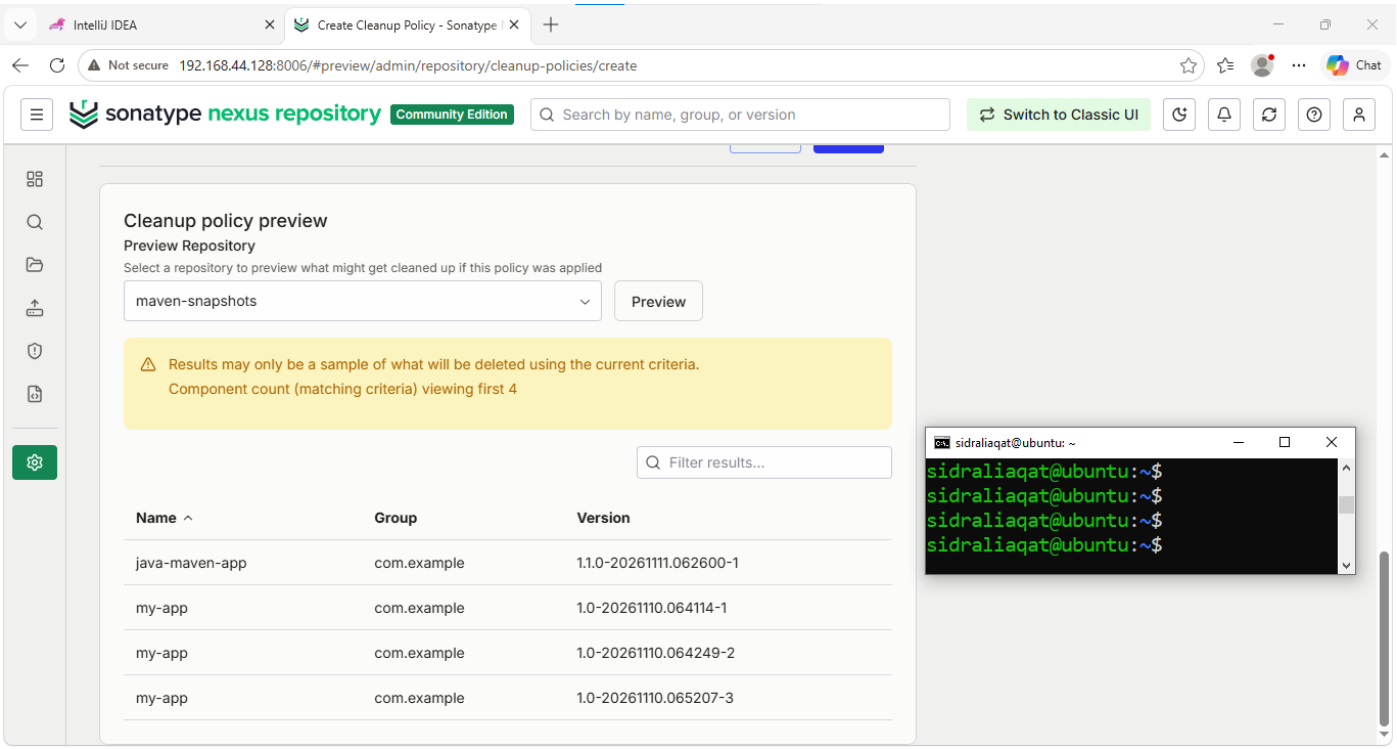


nexus_blob_directories.png
default and mystore directories inside the container shell.

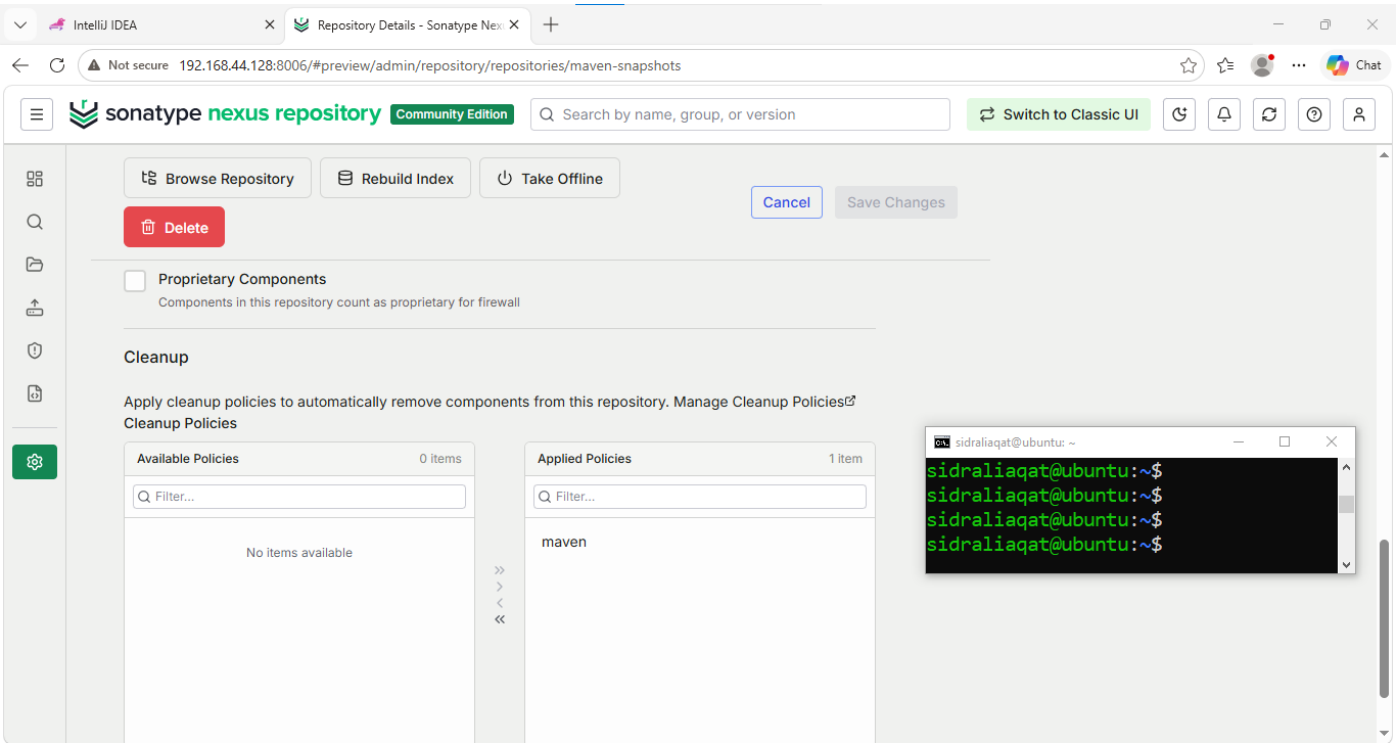


nexus_docker_hosted_repository.png
docker-hosted repository using mystore on HTTP connector 8007.

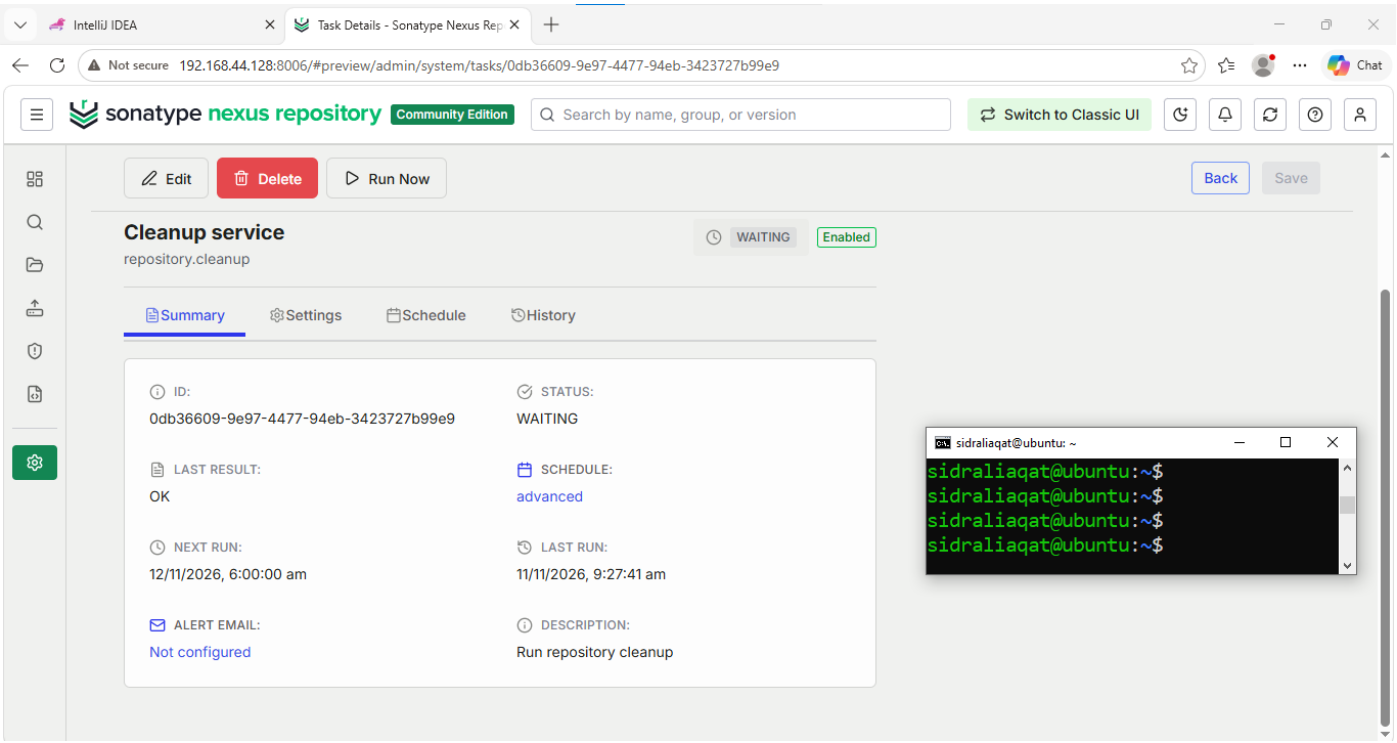
Task 7: Cleanup Policy and Blob Store Compaction



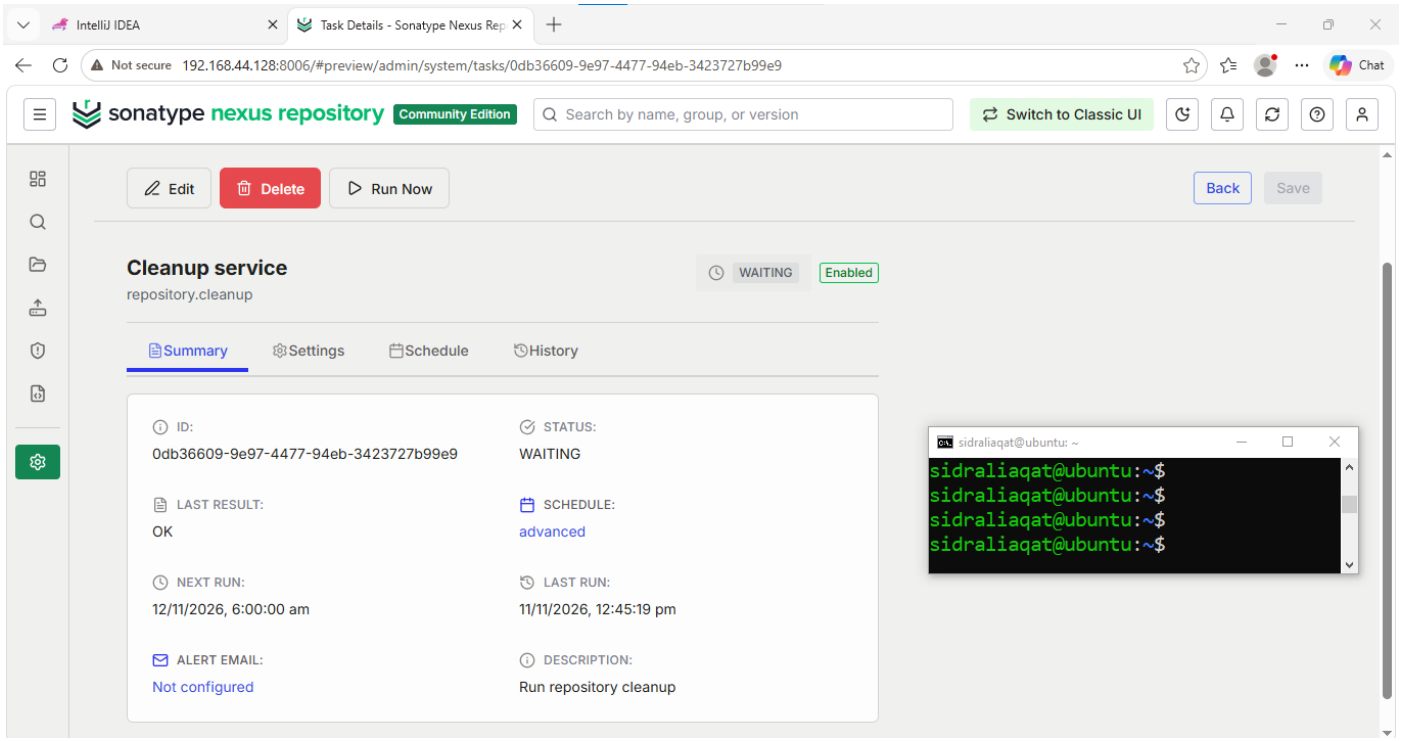
nexus_cleanup_policy_created.png
maven cleanup policy (maven2, snapshots, .*) with the preview of matching components.



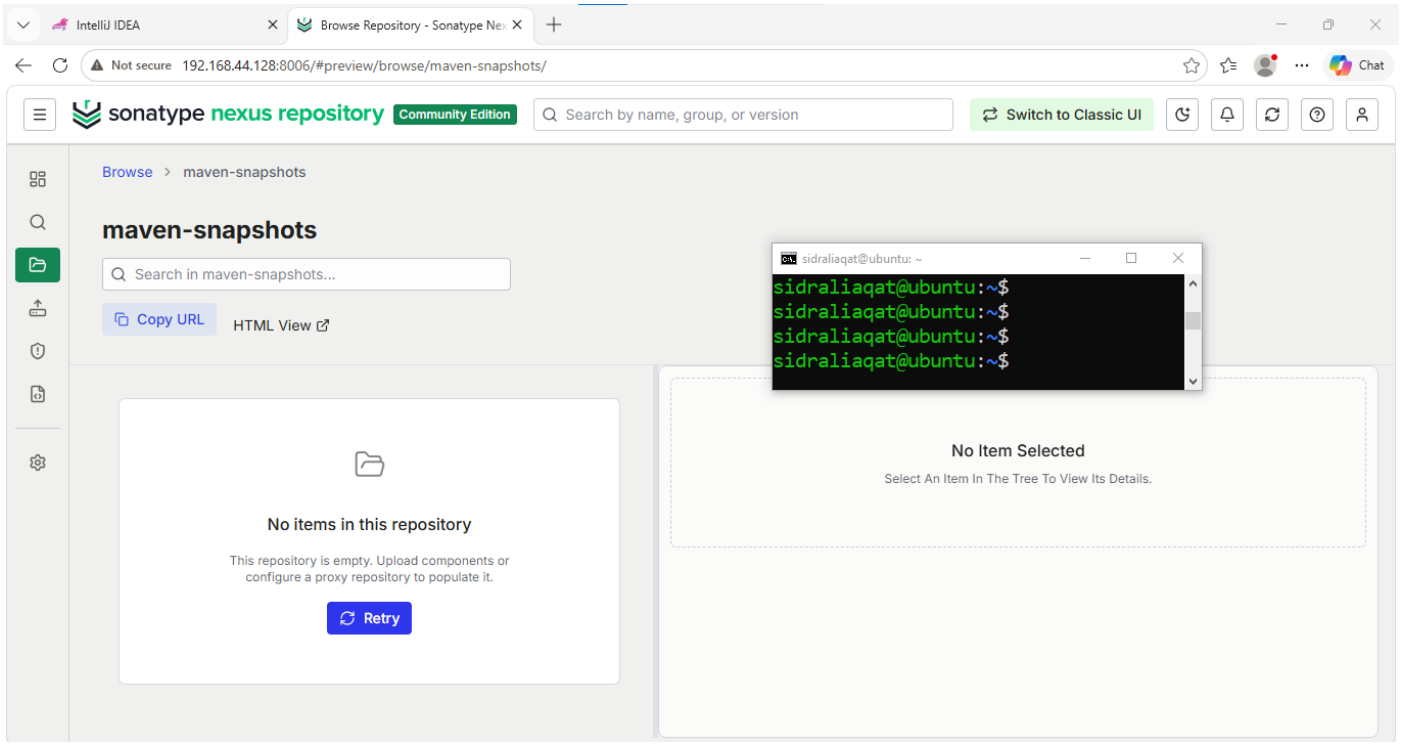
nexus_cleanup_policy_attached.png
maven policy applied to maven-snapshots.



nexus_cleanup_task_waiting.png
Cleanup service in Waiting state, next run at 06:00.



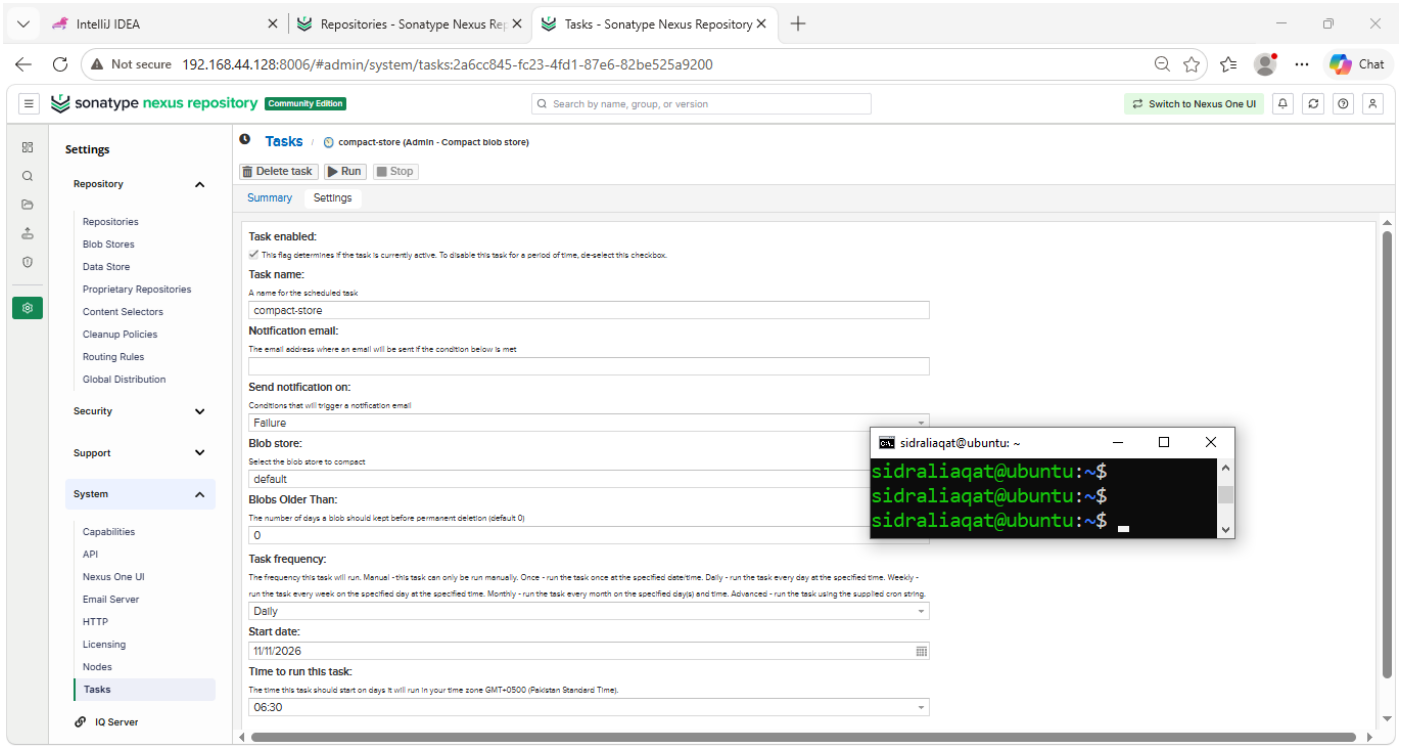
nexus_cleanup_task_completed.png
Cleanup service after the manual run, Last result Ok.



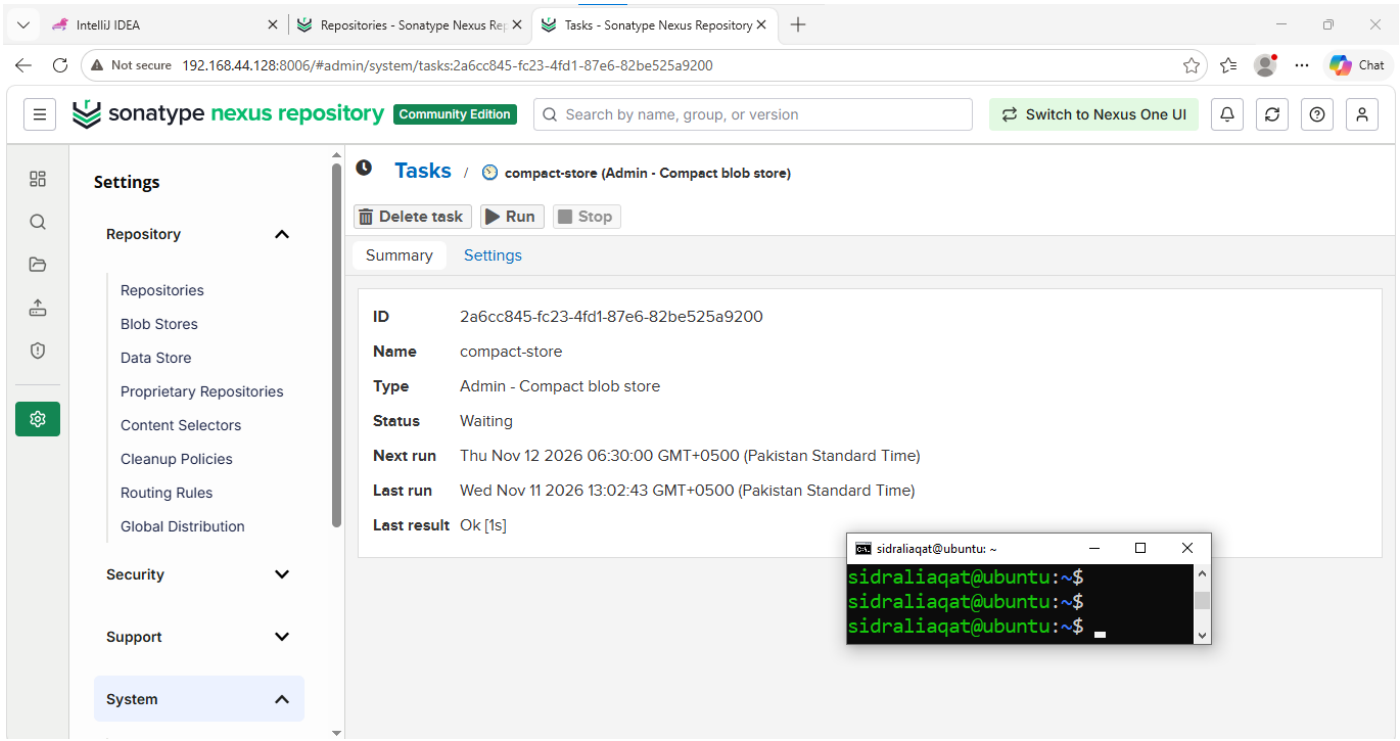
nexus_maven_snapshots_empty.png
maven-snapshots is empty after cleanup.

Screenshot pending: nexus_blob_store_before_compaction.png

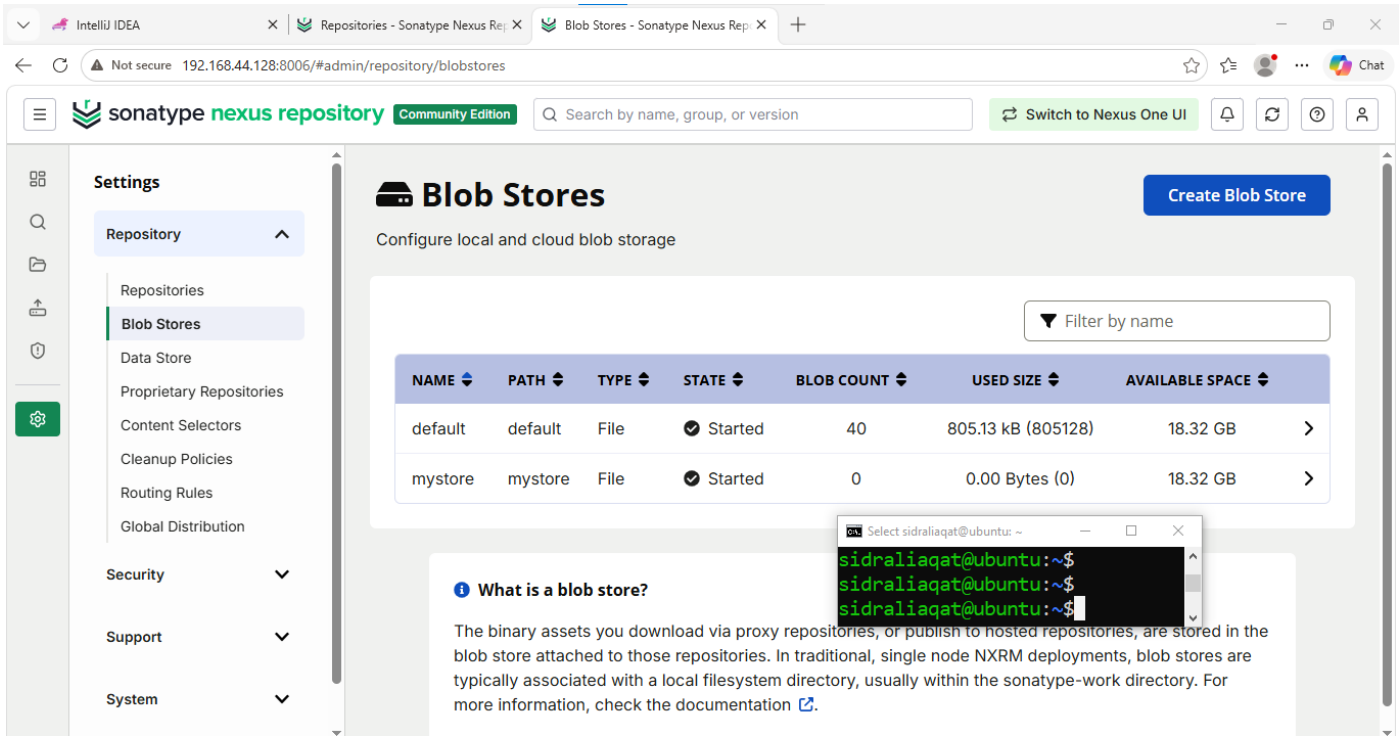
nexus_blob_store_before_compaction.png
Default blob store before compaction: 112 blobs, 158.58 MB.



nexus_compact_task_configured.png
compact-store task: default blob store, 0 days, daily at 06:30.



nexus_compact_task_completed.png
compact-store after the manual run, Last result Ok.



nexus_blob_store_after_compaction.png
Default blob store after compaction: 40 blobs, 805.13 kB.